MATLAB Problem 7.54

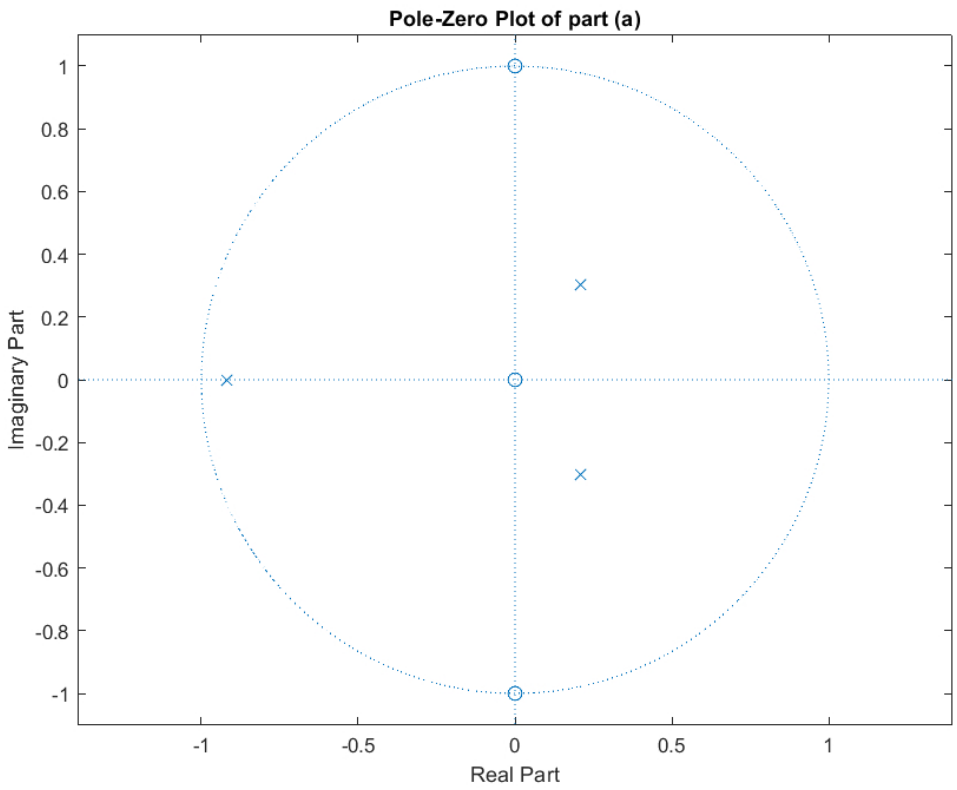
Use the MATLAB command **zplane** to obtain a pole-zero plot for the following systems:

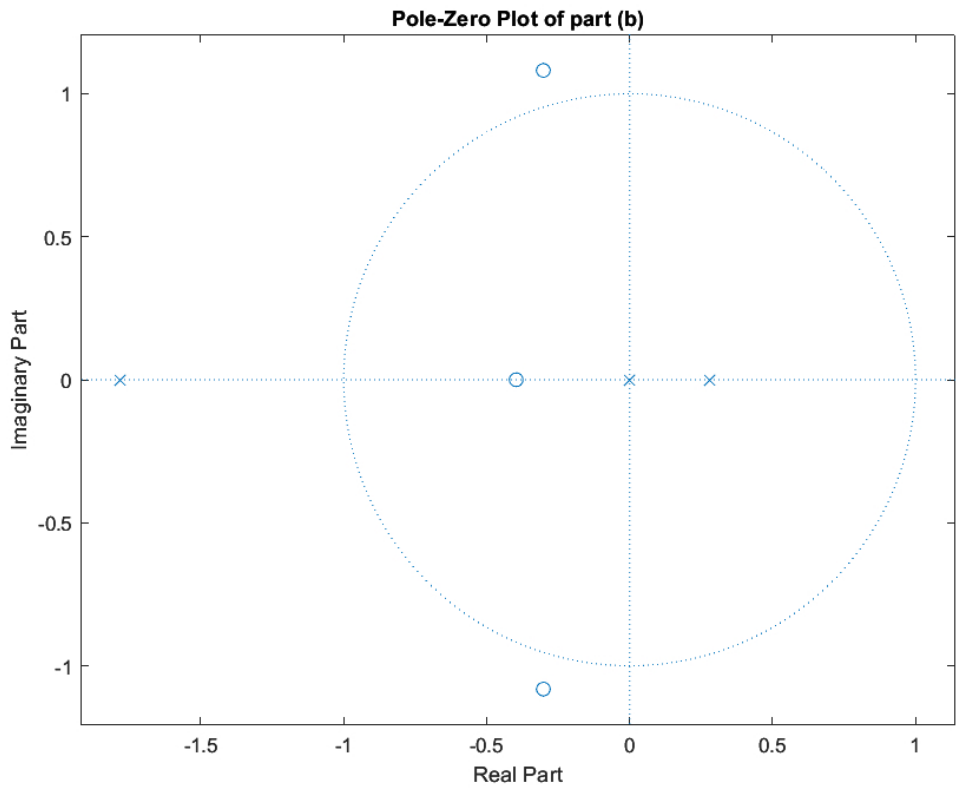
(a) $H(z) = \dfrac{1 + z^{-2}}{2 + z^{-1} - 1/2z^{-2} + 1/4z^{-3}} = \dfrac{z^3 + z}{2z^3 + z^2 - 1/2z + 1/4}$

(b) $H(z) = \dfrac{1 + z^{-1} + 3/2z^{-2} + 1/2z^{-3}}{1 + 3/2z^{-1} - 1/2z^{-2}} = \dfrac{z^3 + z^2 + 3/2z + 1/2}{z^3 + 3/2z^2 - 1/2z}$

```
% (a)
figure(2); clf; clear; clc; format compact
num = [1 0 1 0];
den = [2 1 -1/2 1/4];
zplane(num,den)
title('Pole-Zero Plot of part (a)')

% (b)
figure(2); clf; clear; clc; format compact
num = [1 1 3/2 1/2];
den = [1 3/2 -1/2 0];
zplane(num,den)
title('Pole-Zero Plot of part (b)')
```


Pole-Zero Plot of part (a)
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1


Pole-Zero Plot of part (b)
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1

MATLAB Problem 7.55

Use the MATLAB command **residuez** to obtain the partial-fraction expansions required to solve Problem 7.24(d)-(g).

(d) $X(z) = \dfrac{z^2 - 3z}{z^2 + 3/2z - 1}, 1/2 < |z| < 2$

(e) $X(z) = \dfrac{3z^2 - 1/4z}{z^2 - 16}, |z| > 4$

(f) $X(z) = \dfrac{z^3 + z^2 + 3/2z + 1/2}{z^3 + 3/2z^2 + 1/2z}, |z| < 1/2$

(g) $X(z) = \dfrac{2z^4 - 2z^3 - 2z^2}{z^2 - 1}, |z| > 1$

```
format compact
clc
clear
num=[1 -3]
den=[1 3/2 -1]
[R, P, K] = residuez(num,den)

format compact
clc
clear
num=[3 -1/4]
den=[1 0 -16]
[R, P, K] = residuez(num,den)
```

```
format compact
clc
clear
num=[1 1 3/2 1/2]
den=[1 3/2 1/2]
[R, P, K] = residuez(num,den)

format compact
clc
clear
num=[2 -2 -2]
den=[1 0 -1]
[R, P, K] = residuez(num,den)
```

```
num = 1x2
       1.0000e+000    -3.0000e+000

den = 1x3
       1.0000e+000     1.5000e+000    -1.0000e+000

d_R = 2x1
       2.0000e+000
      -1.0000e+000

d_P = 2x1
      -2.0000e+000
     500.0000e-003

d_K =
     []
```

```
d_K =
     []
num = 1x2
       3.0000e+000  -250.0000e-003

den = 1x3
       1.0000e+000     0.0000e+000   -16.0000e+000

e_R = 2x1
       1.4688e+000
       1.5312e+000

e_P = 2x1
       4.0000e+000
      -4.0000e+000
```

```
num = 1x4
       1.0000e+000     1.0000e+000     1.5000e+000   500.0000e-003

den = 1x3
       1.0000e+000     1.5000e+000   500.0000e-003

f_R = 2x1
       2.0000e+000
      -1.0000e+000

f_P = 2x1
      -1.0000e+000
    -500.0000e-003

f_K = 1x2
       0.0000e+000     1.0000e+000
```

```
num = 1x3
       2.0000e+000    -2.0000e+000    -2.0000e+000

den = 1x3
       1.0000e+000     0.0000e+000    -1.0000e+000

g_R = 2x1
       1.0000e+000
      -1.0000e+000

g_P = 2x1
      -1.0000e+000
       1.0000e+000

g_K =      2.0000e+000
```